

Generation Interconnection System Impact Study Report

For Queue Project Q21-1867

Roanoke Rapids 138 kV

October 29, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1867. The Interconnection Customer proposes a 120.3 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on August 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 625 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$712,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Fluvanna P.S – Bear Substation 345 KV line
Contingency: Leavitt – Birchwood 345 KV
Loading: 105.3% (Rating: 2,071 MVA)
DFAX: 17.4%
Pre-project loading: 104.3%
2. Facility: Natchez – Lakeside 138 KV
Contingency: Mitchell Dam Substation – Skaggs Island Junction 2 345 KV
Loading: 104.1% (Rating: 625 MVA)
DFAX: 6.7%
Pre-project loading: 102.8%
3. Facility: Fritch – Lakeside 138 KV L
Contingency: Skaggs Island Jct 2 – Mitchell Dam 345 KV
Loading: 102.1% (Rating: 706 MVA)
DFAX: 6.7%
Pre-project loading: 101.0%
4. Facility: Skaggs Island Junction 2 – Mitchell Dam 345 KV
Contingency: Bear – Mecklenburg Power Sta Substation 345 KV
Loading: 101.3% (Rating: 2,032 MVA)
DFAX: 27.0%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bear – Fluvanna .S 345 KV L	Reconductor line	\$49,895,000	1% (\$417,000)
Natchez – Lakeside 138 KV	Reconductor line	\$719,000	1% (\$7,000)
Fritch – Lakeside 138 KV L	Reconductor line	\$834,000	1% (\$8,000)
Mitchell Dam – Skaggs Island Junction 2 345 KV	Reconductor line	\$24,440,000	1% (\$280,000)
	Total allocated to project		\$712,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.